JINGXUAN WANG

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Personal Information

Chinese Name: 王景煊 | Major: Software Engineering | English: IELTS 7.0

Research Interests

AI for Drug Discovery | Precision Medicine | Multimodal Learning in Healthcare | Single-cell & Omics Data Analysis | Spatial Transcriptomics | Virtual Cell Modeling & Perturbation Prediction

Education

Beijing University of Posts and Telecommunications (BUPT) Beijing, China
B.Eng. in Software Engineering Sep 2023 - Jun 2027
GPA: 88.68 / 100 ; 34 / 152
Core Courses: Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Data Mining, Mathematical Statistics, Optimization Methods, Database Systems, Software Engineering

Publications

1. **Wang, J.**, Yang, J., Khan, M. A., Yee, P. L., Baili, J., Hu, D. DGAN-MPCC: A Novel Dual-GAN Enhanced Multi-Positive Contrastive Clustering Method for Omics Data. *IEEE Journal of Biomedical and Health Informatics*, 2025. (Accepted, JCR Q1). DOI: 10.1109/JBHI.2025.3631973

2. **Wang, J.**, Guan, R., Liang, K., Wang, S., Yang, X., Cui, X., Hu, D. BiFlow-GAN: Integrating Bidirectional Flows and Cross-Modal Enhancement for Multi-Modal Spatial Transcriptomics. *IEEE Transactions on Multimedia* (Under Review, Submitted Apr 2026, JCR Q1).

3.**Wang, J.**, Guan, R., Xiao, H., Zhang, F., Nan, T., Cui, X., Hu, D. From Representation Learning to Foundation Models: A Survey of Multimodal Fusion in Spatial Transcriptomics and Histopathology. *IEEE Reviews in Biomedical Engineerin*g (Under Review, JCR Q1), 2026.

4. Zhu, S., **Wang, J.**, Hu, X., Tao, J. Enhancing Study-Abroad English Learning with AI-Powered Scenario Simulation. *In Proceedings of ICFULL 2025 & GLoCALL 2025, 2025*. (Oral/Poster Presentation)

Patents & Software Copyright

1. **Wang, J.**, Tao, X. ListenSpeakStar: A Multi-Agent Collaborative Intelligent Training System for English Listening and Speaking. Software Copyright Registration, China Copyright Protection Center, 2025. Reg. No.: 2025SR2109214.

Research Experience

Multi-Modal Spatial Transcriptomics Analysis Nov 2025 - May 2026
- Proposed BiFlow-GAN, a bidirectional-flow generative adversarial framework for robust tissue domain identification in spatial transcriptomics.
- Designed a bidirectional cross-modal attention flow (Bi-AF) mechanism with reciprocal Image-to-Gene (I2G) and Gene-to-Image (G2I) pathways for dynamic feature refinement.
- Benchmarked on diverse ST datasets, significantly outperforming state-of-the-art methods in spatially coherent domain identification and biological interpretability.
- Manuscript submitted to IEEE Transactions on Multimedia in April 2026. (Under Review).

AI-driven Omics Data Clustering & Representation Learning Dec 2024 - Nov 2025
- Proposed a dual-GAN framework to enhance both input and latent representations for noisy single-cell data.
- Designed a multi-positive contrastive clustering strategy to model continuous cell-state transitions.
- Evaluated on multiple real-world single-cell datasets, achieving state-of-the-art clustering performance.
- Published in IEEE Journal of Biomedical and Health Informatics (JCR Q1, 2025).

AI-Powered Scenario Simulation for Language Learning Apr 2025 - Jul 2025
- Conducted an undergraduate innovation research project focused on AI-driven scenario simulation for study-abroad English learning.
- Applied statistical analysis and visualization to evaluate learning outcomes across different simulated scenarios, identifying patterns in learner performance and engagement.
- Resulted in a peer-reviewed conference paper and presentation at ICFULL 2025 & GLoCALL 2025.

ListenSpeakStar: Multi-Agent Collaborative English Training System

May 2025 - Jan 2026

- Developed a multi-agent workflow for English listening and speaking training, integrating scenario generation, learner interaction, error diagnosis, adaptive feedback, and performance evaluation agents. Built a modular orchestration framework to coordinate multi-turn interactions among agents, enabling personalized practice sessions and structured learning feedback.
- Achieved software copyright registration for ListenSpeakStar V1.0. Reg. No.: 2025SR2109214.

Internship

Institute for AI Industry Research (AIR), Tsinghua University

Research Intern — Sep 2026 - Mar 2027

Advisor: Prof. Yanyan Lan (Vice Dean and Chief Researcher, AIR, Tsinghua University)

- Selected as one of 20 trainees for the Chia Tai Tianqing AIDD Future Scholars Research Training Camp at AIR, Tsinghua University (Oct 2026 - Mar 2027).
- Research focus: virtual cell modeling and perturbation-response prediction in spatial transcriptomics, connecting cellular expression states with tissue spatial context.

International Programs & Academic Exchange

SOC Summer Workshop 2025 - National University of Singapore (NUS) May - Jul 2025

- Course: Big Data Analytics and Visualization. Grade: A+. Programme intensity equivalent to a regular semester course at NUS School of Computing.

OxCam Programme 2024 - University of Oxford & University of Cambridge Aug 2024

- AI + Biotechnology Engineering & Healthcare Technology Track. Overall Evaluation Score: A-. Focused on AI-driven healthcare technology and biomedical applications.

Beijing Medical Innovation Center Winter Training Program 2026 Feb 2026

- Selected as participant; focused on medical innovation and translational research.
- Delivered comprehensive presentation on Alzheimer's Disease covering pathological mechanisms.
- Awarded “Outstanding Trainee” for exceptional performance in group presentations and academic discussions.

Competitions & Awards

- China Students Service Outsourcing Innovation & Entrepreneurship Competition (15th Ed.) - National Third Prize, 2024. Cert. No.: F2403243A3293.
- Global Campus AI Algorithm Elite Competition (6th Ed.), Beijing Regional Final, AI + Medicine Track - Second Prize, 2024. Project: Medical Ollama.
- China Students Service Outsourcing Innovation & Entrepreneurship Competition (16th Ed.), Beijing Regional - Provincial Third Prize, 2025. Cert. No.: NT2402245A3068.
- National Undergraduate Statistical Modeling Competition (11th Ed.), Beijing - Regional First Prize, 2025. Project: Spatiotemporal Efficiency Evaluation and Multi-objective Simulation for Medical Resource Allocation. Cert. No.: 20251101A0080.
- China Undergraduate Computer Design Competition (18th Ed.), Beijing Municipal - Regional Second Prize, 2025. Project: Zhang Heng Mathematics AI-Agent. Project ID: 2025003391.
- National Artificial Intelligence Application Innovation Competition, University-Level - Excellence Award, 2025. Team: Fat Orange Thinking.
- University Third-class Scholarship, 2023-2024.
- University Second-class Scholarship, 2025-2026.
- Academic Progress Award, 2025-2026.

Skills

Programming & Development: Python, PyTorch, R, SQL, Git, Linux

Machine Learning & AI: Deep Learning, Contrastive Learning, Generative Models (GANs), Graph Neural Networks (GNN), Vision Transformer (ViT), Representation Learning, RAG, LLMs, Multi-Agent Systems

Biomedical & Data Science: Single-cell & Omics Data Analysis, Spatial Transcriptomics (10x Visium, Stereo-seq), Multimodal Learning in Healthcare, Biomedical Text Mining

Computational Research Methods: Cross-modal feature fusion, cell-state representation learning, clustering evaluation, comparative benchmarking on single-cell and spatial transcriptomics datasets